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Transport device that transports recording medium, and image forming apparatus including same

Abstract

A transport device includes a transport mechanism and a carry in/out mechanism. The transport mechanism transports a recording medium from inside of an image forming apparatus that forms an image on the recording medium, to outside of the image forming apparatus, and transports the recording medium carried in from the outside of the image forming apparatus, to the inside thereof. The carry in/out mechanism performs, in a linked manner, an operation including receiving the recording medium transported by the transport mechanism toward the outside, and further transporting the recording medium toward the outside of the image forming apparatus, and an operation including receiving the recording medium carried in from the outside, and further transporting the recording medium toward the transport mechanism.

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application claims priority to Japanese Patent Application No. 2023-098696 filed on Jun. 15, 2023, the entire contents of which are incorporated by reference herein.

BACKGROUND

(2) The present disclosure relates to a transport device and an image forming apparatus.

(3) A processing unit is known that suppresses a developing unit from wobbling, while maintaining the pressure of a developing roller against a photoconductor drum (photosensitive body) at an appropriate level. Such a processing unit uses a first pressure member and a second pressure member, to maintain the pressure of the developing roller at the appropriate level. In an image forming apparatus including such a processing unit, a processing device that, for example, dries a recording medium on which an image has been formed, may be provided.

SUMMARY

(4) The disclosure proposes further improvement of the foregoing techniques.

(5) In an aspect, the disclosure provides a transport device including a transport mechanism and a carry in/out mechanism. The transport mechanism transports a recording medium from inside of an image forming apparatus that forms an image on the recording medium, to outside of the image forming apparatus, and transports the recording medium carried in from the outside of the image forming apparatus, to the inside thereof. The carry in/out mechanism performs, in a linked manner, an operation including receiving the recording medium transported by the transport mechanism toward the outside, and further transporting the recording medium toward the outside of the image forming apparatus, and an operation including receiving the recording medium carried in from the outside, and further transporting the recording medium toward the transport mechanism.

(6) In another aspect, the disclosure provides an image forming apparatus that forms an image on a recording medium. The image forming apparatus includes the transport device inside the image forming apparatus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic drawing showing an image forming system according to an embodiment of the disclosure;
- (2) FIG. 2 is a schematic drawing showing the carry in/out mechanism shown in FIG. 1;
- (3) FIG. 3 is a schematic drawing showing a drive device that causes a carry-out roller and a carry-in roller to rotate;
- (4) FIG. 4 is a block diagram showing a configuration of a transport device according to the embodiment of the disclosure;
- (5) FIG. 5 is a schematic drawing for explaining an operation of the image forming system according to the embodiment of the disclosure;
- (6) FIG. 6 is another schematic drawing for explaining the operation of the image forming system according to the embodiment of the disclosure, and showing an operation of the transport device, performed when a first detector and a second detector detects a sheet; and
- (7) FIG. 7 is another schematic drawing for explaining the operation of the image forming system according to the embodiment of the disclosure, and showing an operation of the transport device, performed when an abnormality occurs in the image forming apparatus, and the first detector and the second detector no longer detect the sheet.

DETAILED DESCRIPTION

(8) Hereafter, a transport device and an image forming system according to an embodiment of the disclosure will be described, with reference to the drawings. In the drawings, the same or corresponding elements are given the same reference numeral, and the description of such elements will not be repeated.

(9) Referring to FIG. 1, the outline of the image forming system **100** according to the embodiment will be described. FIG. 1 is a schematic drawing showing the image forming system **100** according to the embodiment of the disclosure.

(10) The image forming system **100** includes, as shown in FIG. 1, an image forming apparatus **1**, a postprocessing device **2**, and a transport device **3**.

(11) The image forming apparatus **1** is configured to form an image on a recording medium, either by an electrophotographic method or an ink jet printing method. The image forming apparatus **1** may be, for example, a printer, a copier, or a multifunction peripheral (MFP). The image forming apparatus **1** shown in FIG. 1 represents an electrophotographic printer.

(12) The image forming apparatus **1** forms, on the basis of image data, an image represented by image data on a sheet S (exemplifying the recording medium). The image forming apparatus **1** forms a toner image on the sheet S, for example by the electrophotographic method. Although not

shown, the image forming apparatus **1** may form an image on the sheet S, by the ink jet printing method.

(13) The postprocessing device **2** performs postprocessing on the sheet S, having an image formed thereon. The type of the postprocessing is specified as desired, depending on a position on the sheet S where the image is formed, or user's requirement about the picture quality of the image formed on the sheet S. Examples of the postprocessing include drying, reversing the sheet S, and forming a bundle of a plurality of sheets S.

(14) The transport device **3** transports the sheet S (recording medium), from inside of the image forming apparatus **1**, toward outside thereof. In this embodiment, the transport device **3** transports the sheet S to the postprocessing device **2** on the outside, and receives the sheet S carried in from the postprocessing device **2** on the outside. To be more specific, the transport device **3** transports the sheet S having an image formed thereon by the image forming apparatus **1**, to inside of the postprocessing device **2**. The sheet S transported into the inside of the postprocessing device **2** is again transported by the transport device **3**, from the postprocessing device **2** into the inside of the image forming apparatus **1**. In the subsequent description, the transport of the sheet S from the image forming apparatus **1**, through the postprocessing device **2**, and to the image forming apparatus **1** will be referred to as transport of the sheet S from an upstream side to a downstream side, for the sake of convenience in description. Alternatively, without distinction between the upstream side and the downstream side, the direction in which the sheet S is transported may be referred to as transport direction.

(15) Referring now to FIG. **1** to FIG. **4**, the transport device **3** will be described in detail. FIG. **2** is a schematic drawing showing the carry in/out mechanism **33** shown in FIG. **1**. FIG. **3** is a schematic drawing showing a drive device **334** that causes a carry-out roller **331** and a carry-in roller **333** to rotate. FIG. **4** is a block diagram showing a general configuration of the transport device **3** according to the embodiment of the disclosure.

(16) The transport device **3** includes, as shown in FIG. **1**, a postprocessing transport mechanism **31**, a transport mechanism **32**, a carry in/out mechanism **33**, a first detector **34**, a second detector **35**, and a controller **36**.

(17) As an overview, the postprocessing transport mechanism **31**, the transport mechanism **32**, and the carry in/out mechanism **33** transport the sheet S. The postprocessing transport mechanism **31** is located inside the postprocessing device **2**. The transport mechanism **32** is located inside the image forming apparatus **1**.

(18) The transport mechanism **32** transports the sheet S from the inside of the image forming apparatus **1** toward the postprocessing device **2** on the outside, and transports the sheet S carried in from the postprocessing device **2**, on the outside of the image forming apparatus **1**, into the inside of the image forming apparatus **1**.

(19) The carry in/out mechanism **33** receives and delivers the sheet S, between the postprocessing transport mechanism **31** and the transport mechanism **32**.

(20) The postprocessing transport mechanism **31** includes a first transport unit **311**, and a second transport unit **312**.

(21) The first transport unit **311** receives the sheet S from the carry in/out mechanism **33**. The first transport unit **311** includes a roller **31R** and a belt **31V**. The roller **31R** makes the belt **31V** revolve. The sheet S delivered from the carry in/out mechanism **33** is placed on the revolving belt **31V**. The sheet S placed on the belt **31V** is transported by the revolution thereof, from the upstream side to the downstream side, inside the postprocessing device **2**.

(22) The second transport unit **312** delivers the sheet S transported inside the postprocessing device **2**, to the carry in/out mechanism **33**. The second transport unit **312** includes a pair of rollers **31R**, **31R**. The sheet S is pinched between the pair of rollers **31R**, **31R**. The pinched sheet S is transported by the rotation of the pair of rollers **31R**, **31R**, from the upstream side to the downstream side, inside the postprocessing device **2**.

(23) The transport mechanism **32** transports the sheet S into the inside and toward the outside of the image forming apparatus **1**. The sheet S is delivered to the carry in/out mechanism **33**, from the transport mechanism **32**. In other words, the carry in/out mechanism **33** performs, in a linked manner, an operation including receiving the sheet S transported by the transport mechanism **32** toward the postprocessing device **2** on the outside, and further transporting the sheet S toward the postprocessing device **2** located outside of the image forming apparatus, and an operation including receiving the sheet S carried in from the postprocessing device **2** on the outside, and further transporting the sheet S toward the transport mechanism **32**.

(24) The transport mechanism **32** includes an inward transport unit **321**, an outward transport unit **322**, and a first adsorption device **323**.

(25) The inward transport unit **321** transports the sheet S toward the inside of the image forming apparatus **1**. To be more specific, the inward transport unit **321** transports the sheet S received from the carry in/out mechanism **33**, toward the inside of the image forming apparatus **1**. The inward transport unit **321** includes an inward roller **32A** and an inward belt **32B**. The inward roller **32A** makes the inward belt **32B** revolve. The sheet S delivered from the carry in/out mechanism **33** is placed on the revolving inward belt **32B**. The sheet S placed on the inward belt **32B** is transported by the revolution of the inward belt **32B**, from the upstream side to the downstream side, inside the image forming apparatus **1**.

(26) The outward transport unit **322** transports the sheet S toward the outside of the image forming apparatus **1**. To be more specific, the outward transport unit **322** delivers the sheet S transported inside the image forming apparatus **1**, to the carry in/out mechanism **33**. The outward transport unit **322** includes an outward roller **32C** and an outward belt **32D**. The outward roller **32C** makes the outward belt **32D** revolve. The sheet S is placed on the revolving outward belt **32D**, to be transported inside the image forming apparatus **1**. The outward belt **32D** includes a plurality of small-sized opening **32E**, located with an interval between each other.

(27) The first adsorption device **323** adsorbs the sheet S to the outward transport unit **322**. By doing so, the first adsorption device **323** prevents the sheet S from migrating in an unintended direction, while the sheet S is being transported in the outward transport unit **322**. The first adsorption device **323** may be, for example, a fan that sucks the sheet S toward the outward belt **32D**. The first adsorption device **323** configured as above is located on the opposite side of the sheet S, across the outward belt **32D**. The first adsorption device **323** generates an airflow **F1** that flows from the space where the sheet S is located, toward the first adsorption device **323** through the openings **32E** of the outward belt **32D**. By the airflow **F1** generated by the first adsorption device **323**, the sheet S is adsorbed to the outward belt **32D**.

(28) The carry in/out mechanism **33** delivers the sheet S to the transport mechanism **32** (carry-in operation), which transports the sheet S toward the inside of the image forming apparatus **1**, in linkage with an operation of receiving the sheet S from the transport mechanism **32** (carry-out operation), which transports the sheet S toward the outside of the image forming apparatus **1**. The carry in/out mechanism **33** thus configured performs the carry-in operation and the carry-out operation, using a single drive source. Therefore, the structure of the carry in/out mechanism **33** can be simplified, compared with the case where a plurality of drive sources are required.

(29) To be more specific, as shown in FIG. **1** to FIG. **3**, the carry in/out mechanism **33** includes carry-out rollers **331**, a second adsorption device **332** (see FIG. **2**), a carry-in roller **333**, and a drive device **334** (see FIG. **3**).

(30) The carry-out rollers **331** serve to draw out the sheet S from the transport mechanism **32**. As shown in FIG. **1**, the carry-out rollers **331** are aligned on the downstream side of the outward transport unit **322**, with a clearance that allows the sheet S to stride over. As shown in FIG. **2**, three carry-out rollers **331**, **331**, **331** are aligned along the transport direction, with a clearance **33G**. The sheet S delivered to the carry-out roller **331** is placed on the rotating carry-out roller **331**, to be carried out from the image forming apparatus **1**. The sheet S carried out from the image forming

apparatus **1** is delivered to the first transport unit **311**, by the carry-out roller **331**.

(31) The second adsorption device **332** adsorbs the sheet **S** to the carry-out rollers **331**, as shown in FIG. **2**. By doing so, the second adsorption device **332** prevents the sheet **S**, being transported by the carry-out rollers **331**, from migrating in an unintended direction. To be more specific, the second adsorption device **332** may be, for example, a fan that sucks the sheet **S** toward the carry-out rollers **331**. The second adsorption device **332** is located on the opposite side of the sheet **S**, across the carry-out rollers **331**. The second adsorption device **332** generates an airflow **F2** that flows from the space where the sheet **S** is located, toward the second adsorption device **332** through the clearance **33G**. By the airflow **F2** generated by the second adsorption device **332**, the sheet **S** is adsorbed to the carry-out rollers **331**.

(32) The carry-in roller **333** delivers the sheet **S** to the transport mechanism **32**, as shown in FIG. **1**. To be more specific, the carry-in roller **333** is located on the upstream side of the inward transport unit **321**, with a clearance therefrom. The sheet **S** delivered to the carry-in roller **333** from the second transport unit **312** is delivered to the inward transport unit **321**, by the rotating carry-in roller **333**. As result, the sheet **S** transported by the carry-in roller **333** is delivered to the inside of the image forming apparatus **1**.

(33) The drive device **334** drives the carry-out roller **331** and the carry-in roller **333** in linkage with each other, as shown in FIG. **3**. With such a configuration, the carry-out roller **331** and the carry-in roller **333** are made to rotate by the single drive device **334**. Therefore, the structure of the carry in/out mechanism **33** can be simplified, compared with the case where a plurality of drive sources **334** are required to drive the carry in/out mechanism **33**.

(34) To be more specific, the drive device **334** includes a drive source **33A**, a first transmission unit **33B**, and a second transmission unit **33C**.

(35) The drive source **33A** generates the rotational force for rotating the carry-out roller **331** and the carry-in roller **333**. The drive source **33A** shown in FIG. **3** is a motor having a rotary shaft **33R**.

(36) The first transmission unit **33B** serves to transmit the rotational force of the drive source **33A** to the carry-out rollers **331**. The first transmission unit **33B** shown in FIG. **3** includes a gear train composed of a plurality of gears meshed with each other. The gear train of the first transmission unit **33B** transmits the rotational force of the rotary shaft **33R** to the carry-out roller **331**.

Accordingly, the carry-out roller **331** rotates interlocked with the rotation of the rotary shaft **33R**. Here, the first transmission unit **33B** may have a different structure, without limitation to the gear train, provided that the structure is capable of transmitting the rotational force of the drive source **33A** to the carry-out roller **331**.

(37) The second transmission unit **33C** serves to transmit the rotational force of the drive source **33A** to the carry-in roller **333**. The second transmission unit **33C** shown in FIG. **3** includes, like the first transmission unit **33B**, a gear train composed of a plurality of gears meshed with each other. The gear train of the second transmission unit **33C** transmits the rotational force of the rotary shaft **33R** to the carry-in roller **333**. Accordingly, the carry-in roller **333** rotates interlocked with the rotation of the rotary shaft **33R**. Here, the second transmission unit **33C** may also have a different structure, without limitation to the gear train, provided that the structure is capable of transmitting the rotational force of the drive source **33A** to the carry-in roller **333**.

(38) As described above, the rotational force of the rotary shaft **33R** is transmitted to both of the first transmission unit **33B** and the second transmission unit **33C**. Therefore, the rotation of the rotary shaft **33R** causes the carry-in roller **333** and the carry-out roller **331** to rotate, in linkage with each other.

(39) An abnormality may occur in the image forming apparatus **1** shown in FIG. **1**, during the image forming operation. The abnormality of the image forming apparatus **1** disables the image forming apparatus **1** from continuing with the normal operation. Specific examples of the abnormality of the image forming apparatus **1** include shortage of the sheets **S** on which the image is to be formed, and shortage of ink or toner. The occurrence of such abnormality of the image

forming apparatus **1** is detected by the controller **36**, on the basis of, for example, an output from a sheet sensor provided on the non-illustrated sheet cassette, or a toner sensor provided on a toner cartridge container.

(40) The first detector **34** detects the presence of the sheet **S**, carried out by the carry in/out mechanism **33** from the transport mechanism **32**. To be more detailed, the first detector **34** may be, for example, an optical sensor that detects that at least a part of the sheet **S** has moved away from the outward belt **32D**. As a specific example, the first detector **34** is an optical sensor that detects the presence of the sheet **S**, in the space between the outward belt **32D** and the carry-out roller **331**. Alternatively, the first detector **34** may be constituted of a combination of a position sensor for detecting the presence of the sheet **S** at a desired position, and a processing device that predicts the position of the sheet **S**, on the basis of the detection result from the position sensor and the operation status of each component of the image forming system **100**.

(41) The second detector **35** detects the presence of the sheet **S**, carried in by the carry in/out mechanism **33** to the transport mechanism **32**. To be more detailed, the second detector **35** may be, for example, an optical sensor that detects the presence of the sheet **S** being transported inside the postprocessing device **2**, and the sheet **S** delivered from the carry-in roller **333** to the inward roller **32A**. The second detector **35** may be the combination of the position sensor and the processing unit, described with reference to the first detector **34**. Here, the first detector **34** and the second detector **35** are not limited to the position sensor, or the combination of the position sensor and the processing unit, provided that the sensor is capable of detecting the position of the sheet **S**.

(42) Referring to FIG. **4**, the controller **36** controls the postprocessing transport mechanism **31**, the transport mechanism **32**, and the carry in/out mechanism **33**. Preferably, the controller **36** may control the first adsorption device **323** and the outward transport unit **322**, on the basis of the detection result from the first detector **34**. In addition, the controller **36** controls the drive device **334**, on the basis of the detection result from the second detector **35**. Therefore, the first adsorption device **323**, the outward transport unit **322**, and the drive device **334** each perform an operation appropriate for the position where the sheet **S** is located, when the abnormality occurs in the image forming apparatus **1**. The controller **36** includes a processing device **361** and a storage device **362**.

(43) The controller **36** includes, for example, a processor. The processor may be, for example, a central processing unit (CPU). The controller **36** controls the first transport unit **311**, the second transport unit **312**, the inward transport unit **321**, the outward transport unit **322**, the first adsorption device **323**, the drive device **334**, and the second adsorption device **332**, by executing control programs stored in the storage device **362**. In addition, the controller **36** acts as the processing device **361**, by executing the control program.

(44) The storage device **362** contains the control programs. The storage device **362** is, for example, constituted of a read-only memory (ROM), a random-access memory (RAM), and/or a solid-state drive (SSD), incorporated in the controller **36**. The storage device **362** may include an external memory. The external memory refers to, for example, a removable medium. The storage device **362** may include, for example, a universal serial bus (USB) memory, and/or a secure digital (SD) card, as the external memory.

(45) The function of the controller **36** will be described hereunder, with reference to FIG. **4** to FIG. **7**. FIG. **5** is a schematic drawing for explaining an operation of the image forming system **100** according to the embodiment. FIG. **6** is another schematic drawing for explaining the operation of the image forming system **100** according to the embodiment, and showing an operation of the transport device **3**, performed when the first detector **34** and the second detector **35** have detected the sheet. FIG. **7** is another schematic drawing for explaining the operation of the image forming system **100** according to the embodiment, and showing an operation of the transport device **3**, performed when an abnormality occurs in the image forming apparatus **1**, and the first detector **34** and the second detector **35** do not detect the sheet.

(46) The controller **36** activates the first transport unit **311**, the second transport unit **312**, the

inward transport unit **321**, the outward transport unit **322**, and the drive source **33A**. Accordingly, as shown in FIG. 5, the sheet S transported by the outward transport unit **322** is carried out by the carry-out roller **331**, and delivered to the first transport unit **311**. The sheet S delivered to the first transport unit **311** is transported through inside of the postprocessing device **2**, and then delivered from the second transport unit **312** to the inward transport unit **321**, via the carry-in roller **333**.

(47) In addition, the controller **36** controls the first adsorption device **323** and the second adsorption device **332**, so as to adsorb the sheet S to the outward transport unit **322**, with a force weaker than the force necessary for carrying out the sheet S from the transport mechanism **32**. More specifically, as shown in FIG. 5, the controller **36** controls the first adsorption device **323** and the second adsorption device **332**, such that the air volume of the airflow F1 generated by the first adsorption device **323** becomes smaller than that of the airflow F2 generated by the second adsorption device **332**. As result, the sheet S can be exempted from being subjected to an excessive tensile force, when being delivered from the outward transport unit **322** to the carry-out roller **331**, and prevented from being damaged.

(48) When the sheet S is carried out to the postprocessing device **2**, the controller **36** operates the outward transport unit **322** as shown in FIG. 6. More specifically, the controller **36** controls the rotation of the outward roller **32C**, so as to revolve the outward belt **32D**. Accordingly, the sheet S on the outward belt **32D** is transported toward the outside of the image forming apparatus **1**. With the lapse of time, the sheet S being transported by the outward belt **32D** is carried out from the image forming apparatus **1**. As result, the first detector **34** ceases to detect the sheet S.

(49) The sheet S carried out from the image forming apparatus **1** by the outward belt **32D** is carried into the postprocessing device **2**. In the case where the controller **36** is operating the first transport unit **311**, the second transport unit **312**, and the drive device **334**, the sheet S transported inside the postprocessing device **2** is carried into the carry in/out mechanism **33**, from the second transport unit **312**. As result, the second detector **35** detects the sheet S.

(50) Further, the controller **36** operates the drive device **334**. In other words, the controller **36** causes the drive source **33A** to output the rotational force, so that the carry-out roller **331** and the carry-in roller **333** are made to rotate. Accordingly, the sheet S caught by the carry-in roller **333** is transported toward the outside of the postprocessing device **2**, in other words toward the inside of the image forming apparatus **1**.

(51) When an abnormality occurs in the image forming apparatus **1**, the controller **36** allows the first transport unit **311**, the second transport unit **312**, and the inward transport unit **321** to operate, while the second detector **35** is detecting the sheet S. When the second detector **35** ceases to detect the sheet S, the controller **36** stops the operation of the first transport unit **311**, the second transport unit **312**, and the inward transport unit **321**. Thus, the sheet S located inside the postprocessing device **2** is carried out from the postprocessing device **2** and the carry in/out mechanism **33**. Therefore, when an abnormality occurs in the image forming apparatus **1**, the sheet S being transported does not remain inside the postprocessing device **2**, but is carried into the image forming apparatus **1**.

(52) Referring to FIG. 7, in the case where an abnormality occurs in the image forming apparatus **1**, the controller **36** stops the operation of the outward transport unit **322**, and causes the first adsorption device **323** to operate, when the first detector **34** detects the leading edge of the sheet S. More specifically, the controller **36** stops the rotation of the outward roller **32C**, thereby stopping the revolution of the outward belt **32D**. In addition, the controller **36** causes the first adsorption device **323** to generate the airflow F1. More preferably, the controller **36** may control the drive device **334** so as to stop the rotation of the carry-out roller **331**, and to stop the operation of the second adsorption device **332**. Accordingly, delivery of an additional sheet S to the carry-out roller **331** is prevented. Further, since the first adsorption device **323** generates the airflow F1, the sheet S on the outward transport unit **322** is adsorbed to the outward belt **32D**. As result, the sheet S on the outward belt **32D** is restricted from proceeding toward the postprocessing device **2**.

(53) Further, when an abnormality occurs in the image forming apparatus 1, the controller 36 may stop the outward transport unit 322 and activate the first adsorption device 323 as above, also when the leading edge of the sheet S is not detected by the first detector 34. In this case also, it is more preferable that the controller 36 controls the drive device 334 so as to stop the rotation of the carry-out roller 331, and the operation of the second adsorption device 332.

(54) The mentioned operation allows the sheet S to remain inside the image forming apparatus 1, without being transported to the postprocessing device 2, when an abnormality occurs in the image forming apparatus 1.

(55) In the case of the aforementioned existing image forming apparatus, since the first pressure member and the second pressure member are provided, the mechanism for transporting the recording medium has a complicated structure. In addition, when an error occurs in such image forming apparatus during the image forming operation on the recording medium, it is difficult to remove the sheet from the processing device, and smoothly prevent an additional recording medium from being introduced into the processing device.

(56) According to the foregoing embodiment, in contrast, since the drive device 334 is employed, the sheet S can be transported between the image forming apparatus 1 and the postprocessing device 2, with a simplified structure. According to the embodiment, further, the controller 36 performs the operation as described above, when an abnormality occurs in the image forming apparatus 1. Therefore, even though an error occurs during the image forming operation on the sheet S, the sheet S can be removed from the postprocessing device 2, and an additional sheet S can be smoothly prevented from being carried into the postprocessing device 2.

(57) The embodiment of the disclosure has been described as above, with reference to the drawings. However, the disclosure is not limited to the foregoing embodiment, but may be modified in various manners, without departing from the scope of the disclosure. The drawings each schematically illustrate the elements for the sake of clarity, and the thickness, length, number of pieces, and interval of the illustrated elements may be different from the actual ones, because of the convenience in making up the drawings. Further, the material, shape, and size of the elements referred to in the foregoing embodiment are merely exemplary and not specifically limited, and may be modified as desired, without substantially departing from the configuration according to the disclosure.

INDUSTRIAL APPLICABILITY

(58) The disclosure provides the transport device and the image forming system, and is therefore industrially applicable.

Claims

1. A transport device comprising: a transport mechanism that transports a recording medium from inside of an image forming apparatus that forms an image on the recording medium, to outside of the image forming apparatus, and transports the recording medium carried in from the outside of the image forming apparatus, to the inside thereof; and a carry in/out mechanism that performs, in a linked manner, an operation including receiving the recording medium transported by the transport mechanism toward the outside, and further transporting the recording medium toward the outside of the image forming apparatus, and an operation including receiving the recording medium carried in from the outside, and further transporting the recording medium toward the transport mechanism, wherein the carry in/out mechanism includes: a carry-out roller that receives the recording medium from the transport mechanism and carries out the recording medium; a carry-in roller that receives the recording medium from outside of the image forming apparatus, and carries the recording medium into the transport mechanism; and a drive device that drives the carry-out roller and the carry-in roller in linkage with each other, wherein the transport mechanism includes: an outward transport unit that transports the recording medium toward the outside of the image

forming apparatus, via the carry in/out mechanism; and a first adsorption device that adsorbs the recording medium to the outward transport unit, wherein, when an abnormality occurs in the image forming apparatus, the first adsorption device adsorbs the recording medium to the outward transport unit, with a force greater than a force by which the carry in/out mechanism carries out the recording medium from the transport mechanism.

2. The transport device according to claim 1, further comprising: a second adsorption device provided inside the image forming apparatus at a position downstream of the first adsorption device in a transport direction of the recording medium toward the outside, and configured to adsorb the recording medium to the carry-out roller; and a controller that controls operation of the carry in/out mechanism, the first adsorption device, and the second adsorption device, wherein, when an abnormality occurs in the image forming apparatus, the controller causes the outward transport unit to stop transporting the recording medium, the first adsorption device to operate, and the second adsorption device to stop operating.

3. The transport device according to claim 1, further comprising: a first detector that detects presence of the recording medium being carried out by the carry in/out mechanism from the transport mechanism; a second detector that detects presence of the recording medium being carried by the carry in/out mechanism into the transport mechanism; and a controller that controls the adsorption device and the outward transport unit, on a basis of a detection result from the first detector, and controls the drive device, on a basis of a detection result from the second detector, when an abnormality occurs in the image forming apparatus.

4. The transport device according to claim 3, wherein the controller activates the drive device, when the second detector detects the recording medium.

5. An image forming apparatus that forms an image on a recording medium, the image forming apparatus comprising the transport device according to claim 1, the transport device being incorporated inside the image forming apparatus.
